

4. Lyapunov Stability Theory.

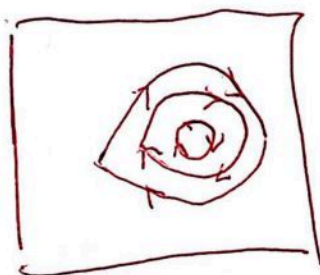
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Lyapunov:

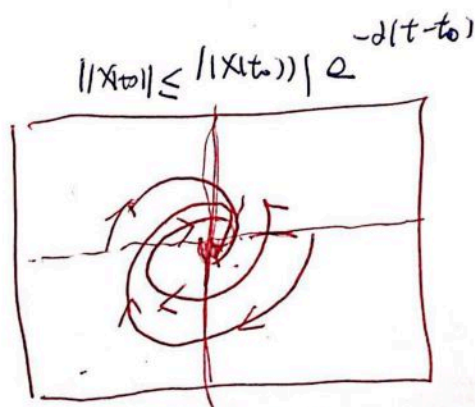
$$\dot{x} = f(x, t), \quad x(t_0) = x_0.$$

L. Stability: $\|x(t_0)\| < \delta \Rightarrow \|x(t)\| < \varepsilon, \quad \forall t \geq t_0$

Asymptotically stable $\|x(t_0)\| < \delta \Rightarrow \lim_{t \rightarrow \infty} x(t) = 0$



L stable.



A stable.

Locally positive definite functions (pdf).

$$V(0, t) = 0$$

$$V(x, t) \geq \alpha(\|x\|), \quad \forall x \in B_\varepsilon = \{x \in \mathbb{R}^n : \|x\| \leq \varepsilon\}$$

(pdf)

$$V(x, t) \geq \alpha(\|x\|), \quad \forall x \in B_\varepsilon, \quad \alpha < \infty$$

Decrescent functions

$$V(x, t) \leq \beta(\|x\|), \quad \forall x \in B_\varepsilon, \quad \forall t \geq t_0$$

$$\dot{V} = \frac{\partial V}{\partial t} + \frac{\partial V}{\partial x} f$$

Table 4.1 Summary of Basic theorems of Lyapunov

4.3. The direct method of Lyapunov.

$$\dot{x} = f(x, t), \quad f(0, t) = 0. \quad (4.37)$$

Define $A = \frac{\partial f(x, t)}{\partial x} \Big|_{x=0}$

$$f_1(x, t) = f(x, t) - A(t)x$$

Assume $\lim_{\|x\| \rightarrow 0} \sup_{t \geq 0} \frac{\|f_1(x, t)\|}{\|x\|} = 0$, then $\dot{z} = A(t)z$, (4.40)

is referred to as the linearization of equation (4.37)

If (4.40) is uniformly asymptotically stable at 0.

then it is locally uniformly asymptotically stable for (4.37).

4.4 Examples.

Example 4.5 Linear Harmonic oscillator.

$$M\ddot{q} + B\dot{q} + Kq = 0 \quad (4.41)$$

$$\frac{d}{dt} \begin{bmatrix} q \\ \dot{q} \end{bmatrix} = \begin{bmatrix} \dot{q} \\ -\frac{K}{M}q - \frac{B}{M}\dot{q} \end{bmatrix} = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} A = \begin{bmatrix} 0 & 1 \\ -\frac{K}{M} & -\frac{B}{M} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\partial f_1}{\partial q} & \frac{\partial f_1}{\partial \dot{q}} \\ \frac{\partial f_2}{\partial q} & \frac{\partial f_2}{\partial \dot{q}} \end{bmatrix}$$

A: characteristic equation

$$\text{real}(\lambda) = \frac{1}{2} \left(-\frac{B}{M} \pm \sqrt{\frac{B^2}{M^2} - 4\frac{KM}{M^2}} \right) < 0. \Rightarrow \text{exponentially stable.}$$

* Lyapunov method.

$$V(x,t) = \frac{1}{2} M \dot{q}^2 + \frac{1}{2} K q^2.$$

$$\dot{V} = M q \dot{q} + K q \dot{q} = -B \dot{q}^2 \leq 0$$

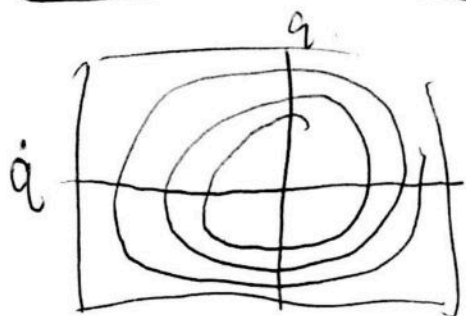
However, it is not locally positive definite.

not depend on q .

$\dot{q} = 0$ at

$$\dot{V} = 0$$

Therefore is not good



$$\tilde{V}(x,t) = \frac{1}{2} \begin{bmatrix} q \\ \dot{q} \end{bmatrix}^T \begin{bmatrix} K & 0 \\ 0 & M \end{bmatrix} \begin{bmatrix} q \\ \dot{q} \end{bmatrix}$$

$$= \frac{1}{2} K q^2 + \frac{1}{2} M \dot{q}^2$$

$$V = \dot{q} M \dot{q} + k q^2 + \epsilon M \dot{q}^2 + \epsilon q M \ddot{q}$$

(4)

$$= (-B + \epsilon M) \dot{q}^2 + \epsilon (-k q^2 - B q \dot{q})$$

$$= - \begin{bmatrix} q \\ \dot{q} \end{bmatrix}^T \begin{bmatrix} \epsilon k & \frac{1}{2} \epsilon B \\ \frac{1}{2} \epsilon B & B - \epsilon M \end{bmatrix} \begin{bmatrix} q \\ \dot{q} \end{bmatrix}$$

$\Rightarrow \dot{V} < 0$ if ϵ sufficient small.

$\epsilon k > 0$.

$$(\epsilon k)(B - \epsilon M) - \frac{1}{4}(\epsilon B)^2$$

$$= \epsilon k B - \epsilon^2 (k M + \frac{1}{4} B^2) > 0 \quad \epsilon \sim 0$$

Example 4.6

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -f(x_2) - g(x_1)$$

$$f(\delta) > 0 \quad \forall \delta \in [-\delta_0, \delta_0] \quad g(\delta) > 0 \quad \forall \delta \in [-\delta_0, \delta_0]$$

$$f(0) = 0 \quad g(0) = 0$$

$$V(x) = \frac{x_2^2}{2} + \int_0^{x_1} g(\delta) d\delta \quad \text{by def.}$$

$$\dot{V} = -x_2 f(x_2) \leq 0 \quad |x_2| \leq \delta$$

4.5. LaSalle invariance principle

$$\dot{x} = f.$$

(5)

① ω limit set. S is the ω limit set of $s(\cdot, x_0, t_0)$ if for every $y \in S$, $\exists \{t_n\}$ such that $s(\cdot, x_0, t_n) \rightarrow y$.

$$n \rightarrow \infty$$

②. Invariant set: $M \subset \mathbb{R}^n$ is said to be an invariant set. if for all $y \in M$ and $t \geq 0$, we have $s(t, y, t_0) \in M$, $\forall t \geq t_0$.

Theorem 4.7. Lasalle's principle.

Let $V: \mathbb{R}^n \rightarrow \mathbb{R}$ be a locally positive definite function.

Such that on the compact set $\Omega_c = \{x \in \mathbb{R}^n : V(x) \leq c\}$, we have

$$\dot{V}(x) \leq 0. \text{ Define } S = \{x \in \Omega_c : \dot{V}(x) = 0\}. \text{ As } t \rightarrow \infty.$$

the trajectory tends to the largest invariant set inside S . If S contain no invariant set other than $x=0$, then 0 is asymptotically stable.

Come back to (4.45)

$$V(x) = \frac{x_2^2}{2} + \int_0^{x_1} g(s) ds \quad \dot{V} = -x_2 f(x_2).$$

$$C = \min |V(-b_0, 0), V(a_0, 0)| \quad \dot{V}(x) \leq 0 \quad \forall x \in \Omega_c.$$

$$\Rightarrow x \rightarrow \text{largest invariant set in } \Omega_c \cap \{x_1, x_2 : \dot{V} = 0\} = \Omega_c \cap \{x_2 = 0\}$$

$$\Rightarrow x_2 = 0 \Rightarrow x_1 \geq x_0 \Rightarrow \dot{x}_2 = 0 = -f(x_2) - g(x_1)$$

$$\Rightarrow g(x_1) = 0 \Rightarrow x_1 = 0 \Rightarrow 0 = \text{Invariant set in } \Omega_c \cap \{x_2 = 0\}$$

$$\dot{V} = 0$$

5. position Control and Trajectory Tracking (b)

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + N(\theta, \dot{\theta}) = \tau. \quad (4.47)$$

design control.

$$\tau = M(\theta)\ddot{\theta}_d - (k_v \dot{e} + k_p e) + C(\theta, \dot{\theta})\dot{\theta} + N(\theta, \dot{\theta}) \quad (4.49)$$

Let (4.47) $\Rightarrow M(\theta)(\ddot{e} + k_v \dot{e} + k_p e) = 0$

$$\Rightarrow \ddot{e} + k_v \dot{e} + k_p e = 0. \quad k_v, k_p \gg 0 \Rightarrow \text{stable}$$

Proposition 48. If k_v, k_p are positive definite, symmetric matrices then the control law applied to (4.47) results in exponentially trajectory tracking.

$$\frac{d}{dt} \begin{bmatrix} e \\ \dot{e} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & I \\ -k_p & -k_v \end{bmatrix}}_A \begin{bmatrix} e \\ \dot{e} \end{bmatrix}$$

Analyze the eigenvalue of A .

$$\lambda \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 & I \\ -k_p & -k_v \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_2 \\ -k_p v_1 - k_v v_2 \end{bmatrix}$$

If $\lambda = 0 \Rightarrow v_2 = 0 \therefore \lambda$ is not an eigenvalue.

If $\lambda \neq 0 \Rightarrow v_2 = 0$ then $\lambda v_1 = 0 \Rightarrow v_1 = 0 \Rightarrow v_1, v_2 = 0$

We can assume $\|v\| = 1$, then we have

$$\begin{aligned} \lambda^2 &= v_1^* \lambda^2 v_1 \xrightarrow{\lambda v_1 = v_2} v_1^* \lambda v_2 \xrightarrow{\lambda v_2 = (-k_p v_1 - k_d v_2)} v_1^* (-k_p v_1 - k_d v_2) \\ &= \underbrace{-v_1^* k_p v_1}_{\beta > 0} - \lambda \underbrace{v_1^* k_d v_2}_{\alpha > 0} \end{aligned} \quad \begin{aligned} &\text{⑦} \\ &\lambda v_2 = (-k_p v_1 - k_d v_2) \\ &\leftarrow v_2 = \lambda v_1 \end{aligned}$$

$$\Rightarrow \lambda^2 + \alpha \lambda + \beta = 0 \quad \alpha, \beta > 0 \Rightarrow \operatorname{Re}(\lambda) < 0.$$

$$\Rightarrow \operatorname{Re}\left(\frac{-\alpha \pm \sqrt{\alpha^2 - 4\beta}}{2}\right) < 0 \rightarrow$$

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5.3. PD Control (proportional plus derivative).

$$\tau = -k_d \dot{e} - k_p e \quad e = \theta - \theta_d. \quad (4.51)$$

P 4.9 If $\dot{\theta}_d \equiv 0$ and $k_d, k_p > 0$, the control law (4.51) applied to (4.47) render the equilibrium $\theta = \theta_d$ globally asymptotically stable.

Proof: For $\dot{\theta}_d \equiv 0$, the closed-loop system is

$$M(\theta) \ddot{\theta} + (C(\theta, \dot{\theta}) \dot{\theta} + k_d \dot{\theta} + k_p (\theta - \theta_d)) = 0 \quad (4.52)$$

~~7.4.2~~ WLOG. we set $\theta_d \equiv 0$ ($\dot{\theta} = \theta - \theta_d$). Then, we define

$$V(\theta, \dot{\theta}) = \frac{1}{2} \dot{\theta}^T M(\theta) \dot{\theta} + \frac{1}{2} \theta^T k_p \theta.$$

V is positive definite and decreases.

$$\begin{aligned} \dot{V} &= \dot{\theta}^T M \ddot{\theta} + \frac{1}{2} \dot{\theta}^T \dot{M} \dot{\theta} + \dot{\theta}^T k_p \theta \\ &= -\dot{\theta}^T k_d \dot{\theta} + \frac{1}{2} \dot{\theta}^T (M - 2C) \dot{\theta}. \end{aligned}$$

Since $(M - 2C)$ is skew-symmetric ^(*), we have $\dot{V} = -\dot{\theta}^T k_d \dot{\theta}$

(*) why? $T = \frac{1}{2} \dot{\theta}^T M \dot{\theta}$ $P = \dot{\theta}^T N$ (8)

$$E = T + P \quad \dot{E} = \dot{T} + \dot{P} = \dot{\theta}^T M \ddot{\theta} + \frac{1}{2} \dot{\theta}^T \dot{M} \dot{\theta} + \dot{\theta}^T N$$

$$\boxed{M \ddot{\theta} + C \dot{\theta} = -N + \tau} = \frac{1}{2} \dot{\theta}^T (M - 2C) \dot{\theta} + \dot{\theta}^T \tau$$

$\therefore \dot{E} = \dot{\theta}^T \tau$ (influence from outside)

$\Rightarrow \dot{\theta}^T (M - 2C) \dot{\theta} = 0 \Rightarrow M - 2C$ skew-symmetric.

Then we use LaSalle's principle.

$$V = 0 \Rightarrow S = \{(\theta, \dot{\theta}) \mid \dot{\theta} = 0\} \quad \underline{\dot{V} = -\dot{\theta}^T K_V \dot{\theta}}$$

Come back to (4.52) $\Rightarrow K_P \theta = 0 \Rightarrow \theta = 0$.

$\Rightarrow$ Largest invariant set contained within S is the single point $\theta = 0$. Hence we finish the proof.

If we consider modified PD control law.

$$\tau = M \ddot{\theta}_d + C \dot{\theta}_d + N - k_v \dot{e} - k_p e \quad (4.53)$$

Then we have

p 4.10. The control law (4.53) applied to (4.67) results in exponential trajectory tracking if $k_v, k_p > 0$.

Proof: The closed-loop system is
$$M \ddot{e} + C \dot{e} + k_v \dot{e} + k_p e = 0 \quad (4.54)$$

Lyapunov x pag. LaSalle x time varying

$$V(e, \dot{e}) = \frac{1}{2} \dot{e}^T M \dot{e} + \frac{1}{2} e^T K_p e + \frac{1}{2} e^T M \omega \dot{e} \quad (9)$$

$$\dot{V} = \dot{e}^T M \ddot{e} + \dot{e} \dot{e}^T M \dot{e} + \dot{e}^T K_p \dot{e} + \frac{1}{2} \dot{e}^T M \dot{e} + \frac{1}{2} \dot{e}^T (M \dot{e} + \dot{M} e)$$

$$= -\dot{e}^T (K_v - \frac{1}{2} M) \dot{e} - \frac{1}{2} \dot{e}^T K_p \dot{e} + \frac{1}{2} \dot{e}^T (-K_v + \frac{1}{2} M) \dot{e}$$

$$= - \begin{bmatrix} \dot{e} \\ e \end{bmatrix}^T \begin{bmatrix} \frac{1}{2} K_p & \frac{1}{2} (K_v - \frac{1}{2} M) \\ \frac{1}{2} (K_v - \frac{1}{2} M) & K_v - \frac{1}{2} M \end{bmatrix} \begin{bmatrix} \dot{e} \\ e \end{bmatrix}$$

$$= - \begin{bmatrix} \dot{e} \\ e \end{bmatrix}^T \begin{bmatrix} \frac{1}{2} K_p & \frac{1}{2} (K_v - \frac{1}{2} M) \\ \frac{1}{2} (K_v - \frac{1}{2} M) & K_v - \frac{1}{2} M \end{bmatrix} \begin{bmatrix} \dot{e} \\ e \end{bmatrix}$$

$\frac{1}{2} K_p > 0$

$$(\frac{1}{2} K_p)(K_v - \frac{1}{2} M) - \frac{1}{4} (K_v - \frac{1}{2} M)^2$$

$$= \frac{1}{2} K_p K_v - \frac{1}{4} [K_p M + (K_v - \frac{1}{2} M)^2]$$

$> 0 \quad (\varepsilon > 0)$

By T4.5

$\Rightarrow$ exponentially stable

How can we get K_v K_p for workspace

θ is angle

$X_d \rightarrow 0$. Complicated.

$X(t) \in SE(3)$ (p, k) PGRS. $0 \in SE(3)$.

5.4 workspace control.

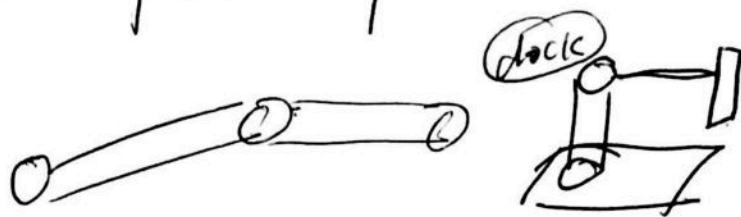
(10)

$$f: Q \rightarrow \mathbb{R}^p.$$

$$0 \in Q, \quad x \in \mathbb{R}^p, \quad (p < 6).$$

partial parameterization.

SCARA robot.



(but we don't want to see f)

$$f(\theta) = x, \quad M(\theta) \ddot{\theta} + C\dot{\theta} + N = \tau.$$

$$\dot{x} = J(\theta) \dot{\theta} \quad J(\theta) = \frac{\partial f}{\partial \theta} \Rightarrow \dot{\theta} = J^{-1} \dot{x} \quad \ddot{\theta} = J^{-1} \ddot{x} + \frac{d}{dt}(J^{-1}) \dot{x}$$

$$J^{-T} = (J^{-1})^T$$

$$J^{-T} M(\theta) J^{-1} \ddot{x} + \left(J^{-T} C J^{-1} + J^{-T} M \frac{d}{dt}(J^{-1}) \right) \dot{x} + J^{-T} N = J^{-T} \tau.$$

We can write this in a more familiar form

$$\tilde{M} = J^{-T} M J^{-1}$$

$$\tilde{C} = J^{-T} \left(C J^{-1} + M \frac{d}{dt}(J^{-1}) \right)$$

$$\tilde{N} = J^{-T} N$$

$$F = J^{-T} \tau.$$

$$\Rightarrow \tilde{M}(\theta) \ddot{x} + \tilde{C} \dot{x} + \tilde{N} = F \quad (4.35)$$

Lemma 4.1. $\tilde{M}$ is symmetric and positive definite

$$\tilde{M} = J^{-T} M J^{-1}$$

2. $\dot{M} - 2\tilde{C}$ is a skew-symmetric matrix. (1)

$$\dot{M} - 2\tilde{C} = \underbrace{J^{-1}(M - 2C)J^{-1}}_{\text{Skew-symmetric}} + \underbrace{\frac{d}{dt} J^{-1} M J^{-1} - J^{-1} \frac{d}{dt} J^{-1}}_{\text{Skew symmetric}}$$

This mean. we can do similar thing to workspace as we do for torques and angles; ~~ie~~

6. Control of constrained Manipulators.

Constrained $h_j(\theta_1, \dots, \theta_n) = 0 \quad j=1, \dots, k \quad (4.86)$

This implies the time derivative $h_j(f_1(\phi), \dots, f_n(\phi)) = 0. \quad (4.87)$

ϕ parameterizes the allowable motion on the surface and $\theta = f(\phi)$ correspond to a configuration θ .

Any torques T_N Normal forces $T_N = \sum \lambda_j \nabla h_j(\theta)$ satisfies

$$T_N \cdot \dot{\theta} = \sum \lambda_i \nabla h_i \cdot \dot{\theta} = \sum \lambda_i \left(\frac{\partial h_i}{\partial \theta} \dot{\theta} \right)$$

$$\Rightarrow \sum \lambda_i \frac{d}{dt} (h_i(\theta)) = 0.$$

Normal force $\leq$ do no work on systems

$\theta = f(\phi)$ Then we can use

$$\dot{\theta} = \frac{\partial f}{\partial \phi} \dot{\phi} \quad \ddot{\theta} = \frac{\partial f}{\partial \phi} \ddot{\phi} + \frac{d}{dt} \left(\frac{\partial f}{\partial \phi} \right) \dot{\phi}$$

$\Rightarrow$ (4.65), (4.121).

more details we left for

Chapter 6: b.

Thank you all!

$$M(\theta) \ddot{\theta} + C(\theta, \dot{\theta}) \dot{\theta} + N(\theta, \dot{\theta}) = \tau$$

$$\Rightarrow M(\theta) \frac{\partial f}{\partial \phi} \ddot{\phi} + \left(C(\theta, \dot{\theta}) \frac{\partial f}{\partial \phi} + M(\theta) \frac{d}{dt} \left(\frac{\partial f}{\partial \phi} \right) \right) \dot{\phi} + N(\theta, \dot{\theta}) = \tau \quad (4.65)$$

$\Rightarrow$ We define

$$\tilde{M} = J^T M(f(\phi)) / J$$

$$\tilde{C}(\phi, \dot{\phi}) = J^T (C(f(\phi), J\dot{\phi}) + M(f(\phi)) \dot{J})$$

$$\tilde{N}(\phi, \dot{\phi}) = J^T (N(f(\phi), J\dot{\phi}))$$

$$\tau = J^T \tau$$

$$\Rightarrow \tilde{M}(\phi) \dot{\phi} + \tilde{C}(\phi, \dot{\phi}) \dot{\phi} + \tilde{N}(\phi, \dot{\phi}) = \tau \quad (4.12)$$

similar to angle ϕ
12'